Contents

25 Supersymmetry Algebras

This chapter will develop the form of the supersymmetry algebra from first principles, following the treatment of Haag, Lopuszanski, and Sohnius [1]. As we shall see, under conditions in which the Coleman–Mandula theorem applies, this structure is almost uniquely fixed by the requirements of Lorentz invariance. The supermultiplet structure of one-particle states will then be deduced directly from the supersymmetry algebra.

25.1 Graded Lie Algebras and Graded Parameters

We saw in Section 2.2 how to express any continuous symmetry transformation in terms of a Lie algebra of linearly independent symmetry generators t_a that satisfy commutation relations $[t_a, t_b] = i\sum_c C^c_{ab} t_c$. In much the same way, supersymmetry is expressed in terms of symmetry generators t_a that form a *graded* Lie algebra [2], embodied in commutation and anticommutation relations of the form

$$t_a t_b - (-1)^{\eta_a \eta_b} t_b t_a = i \sum_c C^c_{ab} t_c \,. \tag{25.1.1}$$

(The summation convention is suspended in this section.) Here η_a for each a is either $+1$ or 0 and is known as the *grading* of the generator t_a, and the C^c_{ab} are a set of numerical structure constants. Generators t_a for which $\eta_a = 1$ are called *fermionic*; the others, for which $\eta_a = 0$, are called *bosonic*. Eq. (25.1.1) provides commutation relations for bosonic operators with each other and with fermionic operators, but anticommutation relations for fermionic operators with each other. We will come back soon to the motivation for Eq. (25.1.1); for the moment we will just take a look at its consequences for the structure constants.

According to Eq. (25.1.1), the structure constants must satisfy the conditions

$$C^c_{ba} = -(-1)^{\eta_a \eta_b} C^c_{ab} \,. \tag{25.1.2}$$

For any operator formed as a functional of field operators, the products of two bosonic or two fermionic operators are bosonic, and the products of a fermionic with a bosonic operator are fermionic, so that

$$C^c_{ab} = 0 \quad \text{unless} \quad \eta^c = \eta^a + \eta^b \pmod{2}\,. \tag{25.1.3}$$

Also, for any operator formed in this way, the Hermitian adjoint of a bosonic or fermionic operator is, respectively, bosonic or fermionic. If the t_a are Hermitian operators, then the structure constants satisfy a reality condition

$$C^{c*}_{ab} = -C^c_{ba}\,. \tag{25.1.4}$$

The structure constants also satisfy a non-linear constraint, which follows from a super-Jacobi identity

$$(-1)^{\eta_c\eta_a}[[t_a, t_b\}, t_c\} + (-1)^{\eta_a\eta_b}[[t_b, t_c\}, t_a\} + (-1)^{\eta_b\eta_c}[[t_c, t_a\}, t_b\} = 0\,. \tag{25.1.5}$$

Here '$[\ldots\}$' denotes a commutator/anticommutator like that appearing on the left-hand side of Eq. (25.1.1), but here extended to any graded operators O, O', etc.

$$[O, O'\} \equiv OO' - (-1)^{\eta(O)\eta(O')}O'O = -(-1)^{\eta(O)\eta(O')}[O', O\}\,, \tag{25.1.6}$$

it now being understood that any product $O = t_a t_b t_c \cdots$ of generators is given a grading $\eta(O) \equiv \eta_a + \eta_b + \eta_c + \cdots \pmod{2}$. (To prove Eq. (25.1.5), it is sufficient to prove that the coefficients of $t_a t_b t_c$ and $t_a t_c t_b$ vanish, for then the symmetry of the left-hand side of Eq. (25.1.5) under the cyclic permutations $abc \longrightarrow bca \longrightarrow cab$ will ensure that the coefficients of all other products of generators also vanish. The coefficient of $t_a t_b t_c$ in Eq. (25.1.5) is

$$(-1)^{\eta_c\eta_a} - (-1)^{\eta_a\eta_b}(-1)^{\eta_a(\eta_b+\eta_c)} = 0\,,$$

while the coefficient of $t_a t_c t_b$ is

$$(-1)^{\eta_a\eta_b}(-1)^{\eta_b\eta_c}(-1)^{\eta_a(\eta_b+\eta_c)} - (-1)^{\eta_b\eta_c}(-1)^{\eta_c\eta_a} = 0\,,$$

completing the proof.) By inserting Eq. (25.1.1) in Eq. (25.1.5), we find the constraint

$$\sum_d (-1)^{\eta_c\eta_a} C^d_{ab} C^e_{dc} + \sum_d (-1)^{\eta_a\eta_b} C^d_{bc} C^e_{da} + \sum_d (-1)^{\eta_b\eta_c} C^d_{ca} C^e_{db} = 0\,. \tag{25.1.7}$$

Of course, in the case where all generators are bosonic Eq. (25.1.5) is the usual Jacobi identity, and Eq. (25.1.7) is the usual non-linear condition (2.2.22) on the structure constants.

Eq. (25.1.1) can be taken as our starting point, but it can also be given a motivation like that given for ordinary Lie algebras in Section 2.2, as a necessary feature of finite continuous symmetry transformations. The difference is that now these transformations depend on continuous *graded* parameters. A set of graded c-number parameters can be thought of as 'numbers,' including Grassmann parameters (see Section 9.5) as well as ordinary numbers, which satisfy the associative and distributive rules of arithmetic, but which instead of simply commuting satisfy relations

$$\alpha^a \beta^b = (-1)^{\eta_a\eta_b}\beta^b\alpha^a\,, \tag{25.1.8}$$

where $\alpha^a, \beta^a, \ldots$ are used to distinguish different values of the ath parameter, in the same way that in vector algebra we might let v^a and u^a denote the a-components of different values of some real vector. Again the ath graded parameter is given a grading η_a equal to $+1$ or 0 when α^a is a fermionic or bosonic parameter, respectively. That is, these parameters commute if either is bosonic, and anticommute if both are fermionic. The product $\alpha^a\beta^b\gamma^c\cdots$ of a set of graded parameters is given the grading $\eta_a + \eta_b + \eta_c + \cdots \pmod 2$; that is, such a product is fermionic if it involves an odd number of fermionic parameters, and is otherwise bosonic. With this grading, it is easy to see that products of graded parameters satisfy a commutation or anticommutation rule just like Eq. (25.1.8).

Consider a continuous transformation $T(\alpha)$, given by a formal power series in the graded parameters α^a:

$$T(\alpha) = \mathbf{1} + \sum_a \alpha^a t_a + \sum_{ab} \alpha^a \alpha^b t_{ab} + \cdots\,, \tag{25.1.9}$$

where t_a, t_{ab}, etc. are a set of α-independent operator coefficients, not yet assumed to satisfy any algebraic relations like Eq. (25.1.1). Because the parameters α^a satisfy Eq. (25.1.8), the coefficients $t_{ab\ldots}$ must satisfy symmetry/antisymmetry conditions, such as

$$t_{ab} = (-1)^{\eta_a\eta_b} t_{ba}\,. \tag{25.1.10}$$

It is convenient also to assume that the transformation $T(\beta)$ commutes with any value α^a of any graded parameter, in which case the operator coefficients in (25.1.9) satisfy the conditions

$$\alpha^a t_b = (-1)^{\eta_a\eta_b} t_b \alpha^a\,, \tag{25.1.11}$$

$$\alpha^a t_{bc} = (-1)^{\eta_a(\eta_b+\eta_c)} t_{bc}\alpha^a\,. \tag{25.1.12}$$

That is, t_b and t_{bc} commute or anticommute with graded parameters as if they were graded parameters themselves, with grading η_b and $\eta_b + \eta_c \pmod 2$, respectively.

The other constraints on these operators follow from the requirement that the $T(\alpha)$ form a semi-group; that is, that the product of the operators for different values α and β of the graded parameters is itself a T-operator

$$T(\alpha)T(\beta) = T\big(f(\alpha,\beta)\big)\,, \tag{25.1.13}$$

where $f^c(\alpha,\beta)$ is itself a formal power series in the graded parameters. Because $T(0)T(\beta) = T(\beta)$ and $T(\alpha)T(0) = T(\alpha)$, we must have

$$f^c(0,\beta) = \beta^c, \qquad f^c(\alpha,0) = \alpha^c\,, \tag{25.1.14}$$

and therefore the power series expansion for $f(\alpha,\beta)$ must take the form

$$f^c(\alpha,\beta) = \alpha^c + \beta^c + \sum_{ab} f^c_{ab}\alpha^a\beta^b + \cdots\,, \tag{25.1.15}$$

where f^c_{ab} is a set of ordinary (that is, bosonic) constants, and '$\cdots$' denotes terms of third or higher order in the graded parameters. In order for $f^c(\alpha,\beta)$ to be a graded parameter, it is necessary for each term in Eq. (25.1.15) to have the same grading, which implies that

$$f^c_{ab} = 0 \quad \text{unless} \quad \eta^c = \eta^a + \eta^b \pmod 2\,. \tag{25.1.16}$$

Inserting the power series (25.1.9) and (25.1.15) into the product rule (25.1.13) gives

$$\left[\mathbf{1} + \sum_a \alpha^a t_a + \sum_{ab} \alpha^a \alpha^b t_{ab} + \cdots\right]\left[\mathbf{1} + \sum_a \beta^a t_a + \sum_{ab} \beta^a \beta^b t_{ab} + \cdots\right]$$
$$= \mathbf{1} + \sum_c \left(\alpha^c + \beta^c + \sum_{ab} f^c_{ab}\alpha^a\beta^b + \cdots\right) t_c$$
$$+ \sum_{cd} \left(\alpha^c + \beta^c + \cdots\right)\left(\alpha^d + \beta^d + \cdots\right) t_{cd} + \cdots .$$

The coefficients of $\mathbf{1}$, α^a, β^a, $\alpha^a\alpha^b$, and $\beta^a\beta^b$ match on both sides of this equation, but the condition that the coefficients of $\alpha^a\beta^b$ are the same yields the non-trivial relation

$$(-1)^{\eta_a\eta_b} t_a t_b = \sum_c f^c_{ab} t_c + t_{ab} + (-1)^{\eta_a\eta_b} t_{ba} = \sum_c f^c_{ab} t_c + 2t_{ab} . \tag{25.1.17}$$

(The *sign* factor on the left-hand side arises from the interchange of t_a and β^b.) Together with higher-order relations of the same sort, this allows us to calculate the whole function (25.1.9) if we know the generators t_a and the group-composition function $f^a(\alpha, \beta)$. But for this to be possible, t_a must satisfy a constraint. Using Eq. (25.1.10), the difference or sum of Eq. (25.1.17) and the same equation with a and b interchanged yields the Lie superalgebra relations Eq. (25.1.1), with structure constants given by

$$iC^c_{ab} = (-1)^{\eta_a\eta_b} f^c_{ab} - f^c_{ba} . \tag{25.1.18}$$

Also, Eq. (25.1.3) follows immediately from Eqs. (25.1.16) and (25.1.18).

The complex conjugate α^* of an anticommuting c-number α is defined so that the Hermitian adjoint of the product of α and an arbitrary operator O is

$$(\alpha O)^\dagger = O^\dagger \alpha^* . \tag{25.1.19}$$

It follows that products of c-numbers behave the same under complex conjugation as operators do under Hermitian conjugation:

$$(\alpha\beta)^* = \beta^*\alpha^* , \tag{25.1.20}$$

and that α^* has the same grading as α.

The graded Lie algebras of importance to physics are severely restricted by spacetime symmetries. We now turn to a consideration of these restrictions.

25.2 Supersymmetry Algebras

Consider a general graded Lie algebra of symmetry generators that commute with the S-matrix. If Q is any of the fermionic symmetry generators, then so will be $U^{-1}(\Lambda)QU(\Lambda)$, where $U(\Lambda)$ is the quantum mechanical operator corresponding to an arbitrary homogeneous Lorentz transformation $\Lambda^\mu{}_\nu$. Therefore $U^{-1}(\Lambda)QU(\Lambda)$ is a linear combination of the complete set of fermionic symmetry generators, and hence this set of generators must furnish a representation of the homogeneous Lorentz group. The individual generators may therefore be classified according to the irreducible representation of the homogeneous Lorentz group to which they belong.

As described in Section 5.6, the representations of the homogeneous Lorentz group furnished by any set of operators can be specified by giving their commutation relations with generators $\mathbf{A}$ and $\mathbf{B}$, defined by

$$\mathbf{A} \equiv \frac{1}{2}(\mathbf{J} + i\mathbf{K}), \qquad \mathbf{B} \equiv \frac{1}{2}(\mathbf{J} - i\mathbf{K}), \tag{25.2.1}$$

where $\mathbf{J}$ and $\mathbf{K}$ are the Hermitian generators of rotations and boosts, respectively. These satisfy the commutation relations

$$[A_i, A_j] = \sum_k \epsilon_{ijk} A_k, \qquad [B_i, B_j] = \sum_k \epsilon_{ijk} B_k, \qquad [A_i, B_j] = 0, \tag{25.2.2}$$

where i, j, and k run over the values $1, 2, 3$, and ϵ_{ijk} is totally antisymmetric, with $\epsilon_{123} = +1$. Thus representations of the homogeneous Lorentz group are labelled, like states with two independent spins, by a pair of integers or half-integers A and B, with elements of the representation labelled by a pair of indices a and b, which run by unit steps from $-A$ to $+A$ and from $-B$ to $+B$, respectively. More specifically, a set of $(2A+1)(2B+1)$ operators Q^{AB}_{ab} that form an (A, B) representation of the homogeneous Lorentz group satisfies the commutation relations

$$[\mathbf{A}, Q^{AB}_{ab}] = -\sum_{a'} \mathbf{J}^{(A)}_{aa'} Q^{AB}_{a'b}\,, \qquad [\mathbf{B}, Q^{AB}_{ab}] = -\sum_{b'} \mathbf{J}^{(B)}_{bb'} Q^{AB}_{ab'}\,, \tag{25.2.3}$$

where $\mathbf{J}^{(j)}$ is the spin three-vector matrix for angular momentum j:

$$\left(J^{(j)}_1 \pm i J^{(j)}_2\right)_{\sigma'\sigma} = \delta_{\sigma',\sigma\pm 1}\sqrt{(j \mp \sigma)(j \pm \sigma + 1)}, \qquad \left(J^{(j)}_3\right)_{\sigma'\sigma} = \delta_{\sigma'\sigma}\sigma. \tag{25.2.4}$$

From Eq. (25.2.4), it follows that[1]

$$-\left(\mathbf{J}^{(j)}\right)^*_{\sigma',\sigma} = (-1)^{\sigma'-\sigma}\left(\mathbf{J}^{(j)}\right)_{-\sigma',-\sigma}. \tag{25.2.5}$$

Thus if Q^j_σ are a set of operators that transform according to the spin j representation of the rotation group, then so are $(-1)^{j-\sigma} Q^{j\dagger}_{-\sigma}$. Also, Eq. (25.2.1) shows that $\mathbf{A}^\dagger = \mathbf{B}$. By taking Hermitian adjoints in Eq. (25.2.3), we see that the Hermitian adjoint $Q^{AB\dagger}_{ab}$ of operators that transform according to the (A, B) representation of the homogeneous Lorentz group are related by a similarity transformation to operators $\overline{Q}^{BA}_{ba}$ that transform according to the (B, A) representation:

$$Q^{AB\dagger}_{ab} = (-1)^{A-a}(-1)^{B-b}\overline{Q}^{BA}_{-b,-a}. \tag{25.2.6}$$

[1] We use a dagger for the Hermitian adjoint of an operator, and an asterisk for the complex conjugate of a number. The dagger $\dagger$ will also be used for the transpose of the matrix formed from the Hermitian adjoints of operators or complex conjugates of numbers.

The Haag–Lopuszanski–Sohnius theorem [1] states in part that the fermion symmetry generators can only belong to the $(0,1/2)$ and $(1/2,0)$ representations. As we have seen, the Hermitian adjoint of a $(0,1/2)$ or $(1/2,0)$ operator is a linear combination respectively of $(1/2,0)$ or $(0,1/2)$ operators, so the complete set of fermionic symmetry operators may be divided into $(0,1/2)$ generators Q_{ar} (with the superscript $0\,\frac{1}{2}$ omitted) and their $(1/2,0)$ Hermitian adjoints $\mathrm{Q}^\dagger_{ar}$, where a is a spinor index running over the values $\pm1/2$, and r is used to distinguish different two-component generators with the same Lorentz transformation properties.[2] The theorem further states that the fermionic generators may be defined so as to satisfy the anticommutation relations

$$\{\mathrm{Q}_{ar}, \mathrm{Q}^\dagger_{bs}\} = 2\delta_{rs}\sigma^\mu_{ab}P_\mu, \tag{25.2.7}$$

$$\{\mathrm{Q}_{ar}, \mathrm{Q}_{bs}\} = e_{ab}Z_{rs}, \tag{25.2.8}$$

where P_μ is the four-momentum operator, the $Z_{rs} = -Z_{sr}$ are bosonic symmetry generators, and σ_μ and e are the $2\cdot 2$ matrices (with rows and columns labelled $+1/2, -1/2$):

$$\begin{gathered}\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \\ \sigma_0 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \qquad e = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}. \end{gathered} \tag{25.2.9}$$

Finally, the fermionic generators commute with energy and momentum:

$$[P_\mu, \mathrm{Q}_{ar}] = [P_\mu, \mathrm{Q}^\dagger_{ar}] = 0, \tag{25.2.10}$$

and the Z_{rs} and $Z^\dagger_{rs}$ are a set of central charges of this algebra, in the sense that

$$\begin{aligned} 0 &= [Z_{rs}, \mathrm{Q}_{at}] = [Z_{rs}, \mathrm{Q}^\dagger_{at}] = [Z_{rs}, Z_{tu}] = [Z_{rs}, Z^\dagger_{tu}] \\ &= [Z^\dagger_{rs}, \mathrm{Q}_{at}] = [Z^\dagger_{rs}, \mathrm{Q}^\dagger_{at}] = [Z^\dagger_{rs}, Z^\dagger_{tu}]. \end{aligned} \tag{25.2.11}$$

To prove these results, let us start by considering non-vanishing fermionic symmetry generators that belong to some (A,B) irreducible representation of the homogeneous Lorentz group, and that therefore may be labelled Q^{AB}_{ab}, where a and b run by unit steps from $-A$ to $+A$ and from $-B$ to $+B$, respectively. As already mentioned, the Hermitian adjoint is related to operators belonging to the (B,A) representation by Eq. (25.2.6), so the anticommutator of these operators must take the form

$$\begin{aligned} \{Q^{AB}_{ab}, Q^{AB\dagger}_{a'b'}\} = (-1)^{A-a'}(-1)^{B-b'} \sum_{C=|A-B|}^{A+B} \sum_{D=|A-B|}^{A+B} \sum_{c=-C}^{C} \sum_{d=-D}^{D} \\ \cdot C_{AB}(Cc; a, -b')C_{AB}(Dd; -a', b)X^{CD}_{cd}, \end{aligned} \tag{25.2.12}$$

[2] We are using roman instead of italic letters for the two-component Weyl spinors Q_{ar}, to distinguish them from four-component Dirac spinors that will be introduced later in this section. There is a notation due to van der Waerden, according to which a $(0,1/2)$ operator like Q is written with dotted indices, as $\mathrm{Q}_{\dot a}$, while $(1/2,0)$ operators are written with undotted indices. We will not use this notation here, but will instead explicitly indicate which two-component spinors transform according to the $(0,1/2)$ or $(1/2,0)$ representations of the homogeneous Lorentz group.

where $C_{AB}(j\sigma;ab)$ is the usual Clebsch–Gordan coefficient for coupling spins A and B to form spin j, and X^{CD}_{cd} is the (c,d)-component of an operator that transforms according to the (C,D) representation of the homogeneous Lorentz group. Using the well-known unitarity properties of the Clebsch–Gordan coefficients, we may express the operator X^{CD}_{cd} in terms of these anticommutators:

$$\begin{aligned} X^{CD}_{cd} = & \sum_{a=-A}^{A}\sum_{b=-B}^{B}\sum_{a'=-A}^{A}\sum_{b'=-B}^{B}(-1)^{A-a'}(-1)^{B-b'} \\ & \cdot C_{AB}(Cc;a,-b')C_{AB}(Dd;-a',b)\{Q^{AB}_{ab},Q^{AB\dagger}_{a'b'}\}. \end{aligned} \tag{25.2.13}$$

Not all of these operators are necessarily non-zero. But the only non-zero Clebsch–Gordan coefficients $C_{AB}(j\sigma,ab)$ for $j=\sigma=A+B$ and $j=-\sigma=A+B$ are the ones for $a=A, b=B$ and $a=-A, b=-B$, respectively, which both have the value unity, so by taking $C=D=c=-d=A+B$ in Eq. (25.2.13), we find that

$$X^{A+B,A+B}_{A+B,-A-B} = (-1)^{2B}\{Q^{AB}_{A,-B},Q^{AB\dagger}_{A,-B}\}. \tag{25.2.14}$$

This cannot vanish unless $Q^{AB}_{A,-B}=0$, which would imply (taking commutators with the 'lowering' operators A_1-iA_2 and 'raising' operators B_1+iB_2) that all the Q^{AB}_{ab} vanish. Hence if there are any non-vanishing (A,B) fermionic generators, then their anticommutators with their adjoints must at least involve non-zero bosonic symmetry generators belonging to the $(A+B,A+B)$ representation.

Now, the Coleman–Mandula theorem tells us that the bosonic symmetry generators consist of the $(1/2,1/2)$ generators P_μ of translations, the $(1,0)+(0,1)$ generators $J_{\mu\nu}$ of proper Lorentz transformations, and perhaps the $(0,0)$ generators T_A of various internal symmetries. (Recall that symmetric traceless tensors of rank N transform according to the representation $(N/2,N/2)$, antisymmetric tensors of rank 2 transform according to the representation $(1,0)+(0,1)$, while Dirac fields transform according to the representation $(1/2,0)+(0,1/2)$.) Hence the fermionic symmetry generators can only belong to representations (A,B) with $A+B\leq 1/2$. These operators turn bosons into fermions and vice-versa, so they cannot be scalars, leaving only the $(1/2,0)$ and $(0,1/2)$ representations, as was to be shown. Labelling the linearly independent $(0,1/2)$ fermionic generators as Q_{ar}, the anticommutator $\{\mathrm{Q}_{ar},\mathrm{Q}^\dagger_{bs}\}$ belongs to the representation $(0,1/2)\cdot(1/2,0)=(1/2,1/2)$, and therefore must be proportional to the only $(1/2,1/2)$ bosonic symmetry generator, the momentum four-vector P_μ. Lorentz invariance dictates that the form of this relation must be

$$\{\mathrm{Q}_{ar},\mathrm{Q}^\dagger_{bs}\} = 2N_{rs}\sigma^\mu_{ab}P_\mu, \tag{25.2.15}$$

where N_{rs} is a numerical matrix.

To see this, we use the isomorphism of the Lorentz group (or more properly its covering group) with the group $SL(2,\mathbb{C})$ of two-dimensional unimodular complex matrices λ, discussed in Section 2.7. The effect of a Lorentz transformation $\Lambda^\mu{}_\nu$ on the $(0,1/2)$ fermionic generators is

$$U^{-1}(\Lambda)\mathrm{Q}_{ar}U(\Lambda) = \sum_b \lambda_{ab}\mathrm{Q}_{br}, \tag{25.2.16}$$

where Λ is the Lorentz transformation defined by

$$\lambda\sigma_\mu\lambda^\dagger = \Lambda^\nu{}_\mu\sigma_\nu. \tag{25.2.17}$$

We can check that Eq. (25.2.16) applies for $(0, 1/2)$ operators by noting that for an infinitesimal Lorentz transformation $\Lambda^\mu{}_\nu = \delta^\mu{}_\nu + \omega^\mu{}_\nu$, with $\omega_{\mu\nu} = -\omega_{\nu\mu}$, Eq. (25.2.17) is satisfied for

$$\lambda = \mathbf{1} + \frac{1}{2}\left[\frac{1}{2}i\epsilon_{ijk}\omega_{ij} + \omega_{k0}\right]\sigma_k,$$

while[3]

$$U(\Lambda) = \mathbf{1} + \frac{1}{2}i\omega_{\mu\nu}J^{\mu\nu} = \mathbf{1} + \frac{1}{2}i\epsilon_{ijk}\omega_{ij}J_k - i\omega_{i0}K_i.$$

(Repeated latin indices i, j, k are summed over values $1, 2, 3$.) In this case, by equating the coefficients of ω_{ij} and ω_{i0} in Eq. (25.2.16), we find that

$$[\mathbf{J}, \mathrm{Q}_a] = -\frac{1}{2}\sum_b \boldsymbol{\sigma}_{ab}\mathrm{Q}_b, \qquad [\mathbf{K}, \mathrm{Q}_a] = -\frac{1}{2}i\sum_b \boldsymbol{\sigma}_{ab}\mathrm{Q}_b,$$

or, equivalently,

$$[\mathbf{B}, \mathrm{Q}_a] = -\frac{1}{2}\sum_b \boldsymbol{\sigma}_{ab}\mathrm{Q}_b, \qquad [\mathbf{A}, \mathrm{Q}_a] = 0,$$

which shows that an operator satisfying Eq. (25.2.16) belongs to the $(0, 1/2)$ representation. Now, the σ_μ form a complete set of $2 \cdot 2$ matrices, so we can put the anticommutator $\{\mathrm{Q}_{ar}, \mathrm{Q}^\dagger_{bs}\}$ in the form $N^\mu_{rs}(\sigma_\mu)_{ab}$, where N^μ_{rs} is some matrix of operators. Eqs. (25.2.16) and (25.2.17) show that these operators are four-vectors, in the sense that $U^{-1}(\Lambda)N^\mu U(\Lambda) = \Lambda^\mu{}_\nu N^\nu$, and so by the Coleman–Mandula theorem they must be proportional to P^μ, the only four-vector of bosonic symmetry operators. Setting $N^\mu_{rs} = 2P^\mu N_{rs}$ then gives Eq. (25.2.15).

Now we will apply a linear transformation to the Q_{ar}, to put their anticommutators in the form (25.2.7). For this purpose, we need to establish that the matrix N_{rs} is Hermitian and positive-definite. That it is Hermitian follows immediately by taking the Hermitian adjoint of Eq. (25.2.15). To see that it is positive-definite, recall that the Q_{ar} are taken linearly independent, so for any non-zero linear combination $Q \equiv \sum_r d_a c_r \mathrm{Q}_{ar}$ there must be some state $\|\Psi\rangle$ that Q does not annihilate. Taking the expectation value of Eq. (25.2.15) in this state implies that

$$2\langle\Psi\|\sum_{ab}\sigma^\mu_{ab}P_\mu d_a d_b^*\|\Psi\rangle\sum_{rs}c_r c_s^* N_{rs} = \langle\Psi\|\{Q, Q^\dagger\}\|\Psi\rangle > 0.$$

This shows immediately that $\sum_{rs} c_r c_s^* N_{rs}$ cannot vanish for any c_r that are not all zero, so N_{rs} is either positive-definite or negative-definite. The operator $\sum_{ab}(\sigma_\mu)_{ab}P^\mu d_a d_b^*$ is positive on the space of physical states with $-P^\mu P_\mu \geq 0$ and $P^0 > 0$, so the matrix N_{rs} must be positive-definite.[4]

[3]Here K_i is defined as J_{i0}. There was a mistake in the first two printings of Volume I: K_i was defined as J^{i0} in Sections 2.4, 3.3, and 3.5, but as J_{i0} in Sections 5.6 and 5.9, with $\mathbf{A}$ and $\mathbf{B}$ throughout given by Eq. (25.2.1).

[4]This argument may be turned around. Assuming a supersymmetry with N_{rs} positive-definite, as in Eq. (25.2.7), we can deduce that $P^0 > 0$ for all states [2]. However, this conclusion is not valid when gravitation is taken into account, and unless gravitation is taken into account a shift in the energy of all states by the same amount would have no physical effects.

We may now define new fermionic generators

$$Q'_{ar} \equiv \sum_s N_{rs}^{-1/2} Q_{as},$$

for which the anticommutators take the form

$$\{Q'_{ar}, Q'^{\dagger}_{bs}\} = 2\delta_{rs}\sigma^{\mu}_{ab}P_{\mu}.$$

From now on we shall assume that the fermionic generators are defined in this way, and drop the primes, so that Eq. (25.2.7) is satisfied.

Next we must show that the Q_{ar} commute with the momentum four-vector P_μ. The commutator of a $(1/2, 1/2)$ operator like P_μ with a $(0, 1/2)$ operator like Q can only be a $(1/2, 0)$ or $(1/2, 1)$ operator, but we have seen that there are no $(1/2, 1)$ symmetry generators, so the commutator of P_μ with Q can only be proportional to the $(1/2, 0)$ symmetry generator $Q^\dagger$. Lorentz invariance requires this relation to take the form

$$[\mathcal{M}_{ab}, Q_{cr}] = \sum_s e_{ac} K_{rs} Q^{\dagger}_{bs}, \tag{25.2.18}$$

where K is a numerical matrix, and $\mathcal{M}$ is the matrix of operators

$$\mathcal{M} \equiv \sigma_\mu P^\mu. \tag{25.2.19}$$

(The matrix e_{ac} is the Clebsch–Gordan coefficient for coupling two spins 1/2 to give zero spin.) It is straightforward then to calculate that

$$\left[\mathcal{M}_{-\frac{1}{2},-\frac{1}{2}}, \left[\mathcal{M}_{-\frac{1}{2},-\frac{1}{2}}, \{Q_{\frac{1}{2}r}, Q^{\dagger}_{\frac{1}{2}s}\}\right]\right] = -4(\mathcal{M})_{-\frac{1}{2},-\frac{1}{2}}(KK^\dagger)_{rs}. \tag{25.2.20}$$

Using Eq. (25.2.7), the left-hand side is a linear combination of multiple commutators $[P_\mu, [P_\nu, P_\lambda]]$, all of which vanish, while $(\mathcal{M})_{-1/2,-1/2}$ is non-zero for generic momenta, so $KK^\dagger = 0$, and therefore $K = 0$, which with Eq. (25.2.18) shows that $[P_\mu, Q_{ar}] = 0$. The Hermitian adjoint gives $[P_\mu, Q^{\dagger}_{ar}] = 0$.

Now we can take up the anticommutator of two Qs. The anticommutator of two $(0, 1/2)$ symmetry generators must be a linear combination of $(0, 1)$ and $(0, 0)$ symmetry generators. The Coleman–Mandula theorem tells us that the only $(0, 1)$ symmetry generators are linear combinations of the generators $J_{\nu\lambda}$ of proper homogeneous Lorentz transformations, but since the Qs commute with P_μ so must their anticommutators, and Eq. (2.4.13) shows that no linear combination of the $J_{\nu\lambda}$ commutes with P_μ. This leaves just $(0, 0)$ operators, which commute with both P_μ and $J_{\mu\nu}$. Lorentz invariance then requires the anticommutators of the Qs with each other to take the form of Eq. (25.2.8). The internal symmetry generators Z_{rs} are antisymmetric in r and s because the whole expression must be symmetric under interchange of r and a with s and b, and the matrix e_{ab} is antisymmetric in a and b.

It now only remains to show that the Zs are central charges. It follows immediately from Eqs. (25.2.8) and (25.2.10) that

$$[P_\mu, Z_{rs}] = 0. \tag{25.2.21}$$

Next consider the generalized Jacobi identity (25.1.5) involving two Qs and a $\mathrm{Q}^\dagger$:

$$0 = [\{\mathrm{Q}_{ar}, \mathrm{Q}_{bs}\}, \mathrm{Q}^\dagger_{ct}] + [\{\mathrm{Q}_{bs}, \mathrm{Q}^\dagger_{ct}\}, \mathrm{Q}_{ar}] + [\{\mathrm{Q}^\dagger_{ct}, \mathrm{Q}_{ar}\}, \mathrm{Q}_{bs}].$$

Eqs. (25.2.7) and (25.2.10) show that the second and third terms vanish, so

$$[Z_{rs}, \mathrm{Q}^\dagger_{ct}] = 0. \tag{25.2.22}$$

Finally, consider the generalized Jacobi identity for a Z, a Q, and a $\mathrm{Q}^\dagger$:

$$0 = -[Z_{rs}, \{\mathrm{Q}_{at}, \mathrm{Q}^\dagger_{bu}\}] + \{\mathrm{Q}^\dagger_{bu}, [Z_{rs}, \mathrm{Q}_{at}]\} - \{\mathrm{Q}_{at}, [\mathrm{Q}^\dagger_{bu}, Z_{rs}]\}.$$

The first and third terms vanish because of Eqs. (25.2.21) and (25.2.22), respectively, so we are left with the second term

$$\{\mathrm{Q}^\dagger_{bu}, [Z_{rs}, \mathrm{Q}_{at}]\} = 0. \tag{25.2.23}$$

Now, $[Z_{rs}, \mathrm{Q}_{at}]$ is a $(0, 1/2)$ symmetry generator, so it must be a linear combination of the Qs:

$$[Z_{rs}, \mathrm{Q}_{at}] = \sum_u M_{rstu} \mathrm{Q}_{au}. \tag{25.2.24}$$

Eq. (25.2.23) then reads

$$\sigma^\mu_{ab} P_\mu M_{rstu} = 0,$$

for all a, b, r, s, t, and u. Since the operator $\sigma^\mu_{ab} P_\mu$ is not zero, we conclude that $M_{rstu} = 0$, so that

$$[Z_{rs}, \mathrm{Q}_{at}] = 0. \tag{25.2.25}$$

Using the anticommutation relation (25.2.8) and its adjoint together with the commutation relations (25.2.22) and (25.2.25) and their adjoints then gives

$$[Z_{rs}, Z_{tu}] = [Z_{rs}, Z^\dagger_{tu}] = [Z^\dagger_{rs}, Z^\dagger_{tu}] = 0, \tag{25.2.26}$$

which finishes the proof of Eq. (25.2.11), and with it the proof of the Haag–Lopuszanski–Sohnius theorem.

Of course, the fact that the Z_{rs} are central charges of the supersymmetry algebra does not rule out the possibility that there may be *other* Abelian or non-Abelian internal symmetries. Let T_A span the complete Lie algebra of bosonic internal symmetries. Then $[T_A, \mathrm{Q}_{ar}]$ is a $(0, 1/2)$ symmetry generator, so it must be a linear combination of the Qs:

$$[T_A, \mathrm{Q}_{ar}] = -\sum_s (t_A)_{rs} \mathrm{Q}_{as}. \tag{25.2.27}$$

From the Jacobi identity for two Ts and a Q, we learn that the t_A matrices furnish a representation of the internal symmetry algebra

$$[t_A, t_B] = i \sum_C C^C_{AB} t_C, \tag{25.2.28}$$

where the coefficients C^C_{AB} are the structure constants of the internal symmetry algebra

$$[T_A, T_B] = i \sum_C C^C_{AB} T_C. \tag{25.2.29}$$

The Z_{rs} will then be central charges not only of the superalgebra consisting of the Qs, $\mathrm{Q}^\dagger$s, P_μ, Zs, and $Z^\dagger$s, but also of the larger symmetry superalgebra that in addition contains all the T_A. To see this, note from Eqs. (25.2.27) and (25.2.8) that

$$[T_A, Z_{rs}] = -\sum_{r'} (t_A)_{rr'} Z_{r's} - \sum_{s'} (t_A)_{ss'} Z_{rs'},$$

so the Z_{rs} form an *invariant* Abelian subalgebra of the whole bosonic symmetry algebra. But recall that in proving the Coleman–Mandula theorem we found that the complete Lie algebra of internal bosonic symmetries, which in our case is spanned by the T_A, is isomorphic to a direct sum of a compact semi-simple Lie algebra and several $U(1)$ algebras. The only invariant Abelian subalgebras of such a Lie algebra are spanned by $U(1)$ generators, so the Z_{rs} must be $U(1)$ generators, and hence commute with all the T_A.

Even though the Zs commute with all symmetry generators, they are not just numbers; they are quantum operators, whose value may vary from state to state. In fact, the Zs must obviously take the value zero for a supersymmetric vacuum state, which is annihilated by all supersymmetry generators, but they need not vanish in general. In Section 27.9 we will see how the Zs may be calculated in gauge theories with extended supersymmetry.

In the absence of central charges, the supersymmetry algebra (25.2.7), (25.2.8) is invariant under a group $U(N)$ of internal symmetries

$$\mathrm{Q}_{ar} \longrightarrow \sum_s V_{rs} \mathrm{Q}_{as}, \tag{25.2.30}$$

with V_{rs} an $N \cdot N$ unitary (not necessarily unimodular) matrix. This is known as R-symmetry. This symmetry may or may not be a good symmetry of the action, and if it is then it may be violated by anomalies, or it may be spontaneously broken, or it may be a good symmetry of nature.

A supersymmetry algebra where r, s, etc. run over $N > 1$ values is known as an N-*extended supersymmetry*. Where there is just one Q, the condition $Z_{rs} = -Z_{sr}$ tells us that the Zs vanish, yielding a simpler form of the anticommutation relations

$$\{\mathrm{Q}_a, \mathrm{Q}^\dagger_b\} = 2\sigma^\mu_{ab} P_\mu, \tag{25.2.31}$$

$$\{\mathrm{Q}_a, \mathrm{Q}_b\} = 0. \tag{25.2.32}$$

This is known as the case of *simple supersymmetry*, or $N = 1$ supersymmetry. In this case the R-symmetry transformations are $U(1)$ phase transformations

$$\mathrm{Q}_a \longrightarrow \exp(i\varphi)\mathrm{Q}_a, \tag{25.2.33}$$

with φ a real phase.

For several purposes it is convenient to combine the $(0,1/2)$ operators Q_{ar} together with $(1/2,0)$ operators, which according to Eq. (25.2.6) may be taken as $e_{ab}\mathrm{Q}^\dagger_{br}$, into four-component Majorana spinor generators $Q_{\alpha r}$, defined as

$$Q_r \equiv \begin{pmatrix} e\mathrm{Q}^\dagger_r \\ \mathrm{Q}_r \end{pmatrix}, \tag{25.2.34}$$

or more explicitly

$$Q_{1r} = \mathrm{Q}^\dagger_{-\frac{1}{2}r}, \qquad Q_{2r} = -\mathrm{Q}^\dagger_{\frac{1}{2}r}, \qquad Q_{3r} = \mathrm{Q}_{\frac{1}{2}r}, \qquad Q_{4r} = \mathrm{Q}_{-\frac{1}{2}r}.$$

This is a Majorana spinor in the sense that

$$Q_r = -\beta\epsilon\gamma_5 Q^*_r,$$

where β, ϵ, and γ_5 are $4\cdot 4$ matrices that may be written as the $2\cdot 2$ block matrices:

$$\beta = \begin{pmatrix} 0 & \mathbf{1} \\ \mathbf{1} & 0 \end{pmatrix}, \qquad \epsilon = \begin{pmatrix} e & 0 \\ 0 & e \end{pmatrix}, \qquad \gamma_5 = \begin{pmatrix} \mathbf{1} & 0 \\ 0 & -\mathbf{1} \end{pmatrix}.$$

(The properties of Majorana spinors are reviewed in the appendix to Chapter 26.) The form (25.2.34) is chosen in accordance with the usual notation for the four-component Dirac representation of the homogeneous Lorentz group, in which according to Eq. (5.4.4) the rotation and boost generators are represented according to Eqs. (5.4.19) and (5.4.20) by

$$\mathcal{J}_i = \frac{1}{2}\begin{bmatrix} \sigma_i & 0 \\ 0 & \sigma_i \end{bmatrix}, \qquad \mathcal{K}_i = -\frac{i}{2}\begin{bmatrix} \sigma_i & 0 \\ 0 & -\sigma_i \end{bmatrix}. \tag{25.2.35}$$

With Eq. (25.2.1), this shows that the operators $\mathbf{A}$ and $\mathbf{B}$ act only on the top two and bottom two components of the Dirac spinor, respectively, which is why we use the $(0,1/2)$ operators Q_{ar} as the bottom rather than the top components in Eq. (25.2.34).

In this four-component notation, the fundamental anticommutation relations (25.2.31) and (25.2.32) for simple supersymmetry read

$$\{Q,\overline{Q}\} = 2\begin{pmatrix} 0 & -e(\sigma_\mu P^\mu)^T e \\ \sigma_\mu P^\mu & 0 \end{pmatrix} = -2iP_\mu\gamma^\mu. \tag{25.2.36}$$

Our notation for Dirac matrices is reviewed in the Preface to this volume; here we need only recall that

$$\gamma^0 = -i\beta = -i\begin{pmatrix} 0 & \sigma_0 \\ \sigma_0 & 0 \end{pmatrix}, \qquad \boldsymbol{\gamma} = -i\begin{pmatrix} 0 & \boldsymbol{\sigma} \\ -\boldsymbol{\sigma} & 0 \end{pmatrix}, \tag{25.2.37}$$

while $e\boldsymbol{\sigma}^T e = \boldsymbol{\sigma}$, $e\sigma_0 e = -\sigma_0$, and as usual $\overline{Q} = Q^\dagger\beta$. The presence of central charges will change this formula in the case of extended supersymmetry; instead of Eq. (25.2.36), we have

$$\{Q_r,\overline{Q}_s\} = -2iP_\mu\gamma^\mu\delta_{rs} + \left(\frac{\mathbf{1}+\gamma_5}{2}\right)Z^\dagger_{sr} + \left(\frac{\mathbf{1}-\gamma_5}{2}\right)Z_{rs}. \tag{25.2.38}$$

The analysis presented here for the case of four spacetime dimensions will be repeated, in a somewhat less explicit form, for general spacetime dimensions in Chapter 32. As we will see there, the supersymmetry generators always belong to the fundamental spinor representation of the higher-dimensional Lorentz group, even in theories with extended objects that allow the construction of bosonic symmetry generators other than those allowed by the Coleman–Mandula theorem.

* * *

In theories of massless particles that are invariant under the conformal symmetry algebra (**??**)–(**??**) there are two additional bosonic symmetry generators D and K_μ that can appear on the right-hand side of the supersymmetry anticommutation relations. These new generators have the Lorentz transformation properties of a scalar and a vector, respectively, just like Z_{rs} and P_μ, so once again the fermionic generators must belong to the fundamental $(1/2, 0)$ spinor representation of the Lorentz algebra, and its Hermitian adjoint, the $(0, 1/2)$ representation. It is convenient to classify all generators also according to their commutators with the dilation generator D; an operator X is said to have dimensionality a if

$$[X, D] = iaX. \tag{25.2.39}$$

Inspection of Eq. (**??**) shows that the bosonic symmetry generators $J^{\mu\nu}$, P^μ, K^μ, and D have dimensionalities $0, +1, -1$, and 0, respectively. Also, the generators of any Lie group of internal symmetries have dimensionality 0. The anticommutator of a fermionic generator of dimensionality a with its adjoint is a positive-definite bosonic operator of dimensionality $2a$, so since the only positive-definite bosonic symmetry generators are linear combinations of the components of P_μ and K_μ, the only fermionic symmetry generators have dimensionalities $+1/2$ and $-1/2$.

The $(0, 1/2)$ fermionic symmetry generators of dimensionality $1/2$ and their adjoints may again be assembled into Majorana spinors $Q_{r\alpha}$, with

$$\{Q_{r\alpha}, \overline{Q}_{s\beta}\} = -2iP_\mu(\gamma^\mu)_{\alpha\beta}\delta_{rs}, \tag{25.2.40}$$

$$[P_\mu, Q_{r\alpha}] = 0, \tag{25.2.41}$$

$$[D, Q_{r\alpha}] = -\frac{1}{2}iQ_{r\alpha}. \tag{25.2.42}$$

(Note that central charges are not allowed here, because their dimensionality would be 0, not $+1$.) The commutators of the K_μ with the $Q_{r\alpha}$ are linear combinations of Majorana fermionic symmetry generators $Q^\#_{r\alpha}$, which Lorentz invariance allows us to write in the form

$$[K^\mu, Q_{r\alpha}] = i(\gamma^\mu)_{\alpha\beta}Q^\#_{r\beta}. \tag{25.2.43}$$

(An arbitrary factor on the right-hand side has been absorbed into the normalization of $Q^\#_{r\beta}$. The phase of the right-hand side is chosen so that $Q^\#_{r\beta}$ will satisfy the standard reality condition (26.A.2) for Majorana spinors.) The $Q^\#_{r\beta}$ have dimensionality $+1/2 - 1 = -1/2$, so

$$[D, Q^\#_{r\alpha}] = +\frac{1}{2}iQ^\#_{r\alpha}. \tag{25.2.44}$$

Taking the commutator of Eq. (25.2.43) with P^ν and using the commutator of K^μ with P^ν given by Eq. (**??**) gives

$$[P^\nu, Q^{\#}_{r\alpha}] = -i(\gamma^\nu)_{\alpha\beta} Q_{r\beta}. \tag{25.2.45}$$

We see that the Qs and $Q^{\#}$s are paired. By taking the commutator of the anticommutation relation (25.2.40) with K_μ we find the anticommutators of the $Q^{\#}$s and Qs:

$$\begin{aligned}\{Q^{\#}_{r\alpha}, \overline{Q}_{s\beta}\} = 2iD\delta_{rs}\delta_{\alpha\beta} + 2J_{\mu\nu}\delta_{rs}\mathcal{J}^{\mu\nu}_{\alpha\beta} \\ + O_{rs}\delta_{\alpha\beta} + O'_{rs}(\gamma_5)_{\alpha\beta},\end{aligned} \tag{25.2.46}$$

where $\mathcal{J}^{\mu\nu} \equiv -i[\gamma^\mu, \gamma^\nu]/4$, and O_{rs} and O'_{rs} are Lorentz invariant operators of zero dimensionality with

$$O_{rs} = -O_{sr}, \qquad O'_{rs} = +O'_{sr}. \tag{25.2.47}$$

Taking the commutator of Eq. (25.2.43) with K_ν and using the fact that $[K_\nu, K_\mu] = 0$, we find that $(\gamma^\mu)_{\alpha\beta}[K^\nu, Q_{r\beta}]$ is symmetric in μ and ν, which with a little algebra tells us that

$$[K^\nu, Q_{r\beta}] = 0. \tag{25.2.48}$$

Also, taking the commutator of Eq. (25.2.46) with K_ν gives

$$\{Q^{\#}_{r\alpha}, \overline{Q^{\#}}_{s\beta}\} = +2iK_\mu(\gamma^\mu)_{\alpha\beta}\delta_{rs}. \tag{25.2.49}$$

Finally, taking the commutator of Eq. (25.2.46) with $Q_{t\gamma}$ shows that O_{rs} and O'_{rs} act as the generators of the R-symmetry group $U(N)$, with the left- and right-handed parts of $Q_{r\alpha}$ transforming according to the representations $\mathbf{N}$ and $\overline{\mathbf{N}}$, respectively, while the P_μ, K_μ and D are $U(N)$-invariant. The $U(N)$ commutation relations of these generators with each other and with other generators together with Eqs. (**??**), (**??**), (25.2.40)–(25.2.49) and the commutators of $J_{\mu\nu}$ and D with the various generators constitute the *superconformal algebra*. One of the outstanding differences between this algebra and that of ordinary simple or N-extended supersymmetry is that the $U(N)$ symmetry is not merely an outer automorphism of the algebra that may or may not be a symmetry of the action—it is a part of the superconformal algebra, which therefore must be a symmetry of any action that is supersymmetric and conformally invariant.

25.3 Space Inversion Properties of Supersymmetry Generators

In theories that respect the conservation of parity, the result $P^{-1}\mathrm{Q}_{ar}P$ of acting on the fermionic symmetry generator Q_{ar} with the parity operator P must also be a fermionic symmetry generator. Since J_i and K_i are respectively even and odd under space inversion, Eq. (25.2.1) shows that the effect of acting on A_i with the parity operator is

$$P^{-1}A_iP = B_i. \tag{25.3.1}$$

According to Eq. (25.2.3), the definition of Q_{ar} as a $(0, 1/2)$ operator means that

$$[B_i, \mathrm{Q}_{ar}] = -\frac{1}{2}\sum_b (\sigma_i)_{ab}\mathrm{Q}_{br}, \qquad [A_i, \mathrm{Q}_{ar}] = 0. \tag{25.3.2}$$

Applying the parity operator yields

$$[A_i, P^{-1}\mathrm{Q}_{ar}P] = -\frac{1}{2}\sum_b (\sigma_i)_{ab}P^{-1}\mathrm{Q}_{br}P, \quad [B_i, P^{-1}\mathrm{Q}_{ar}P] = 0, \tag{25.3.3}$$

so $P^{-1}\mathrm{Q}_{ar}P$ is a $(1/2, 0)$ symmetry generator, and therefore must be a linear combination of the $\mathrm{Q}^\dagger_{ar}$. According to Eq. (25.2.6), Lorentz invariance dictates the form of this relation as

$$P^{-1}\mathrm{Q}_{ar}P = \sum_{bs} \mathcal{P}_{rs}e_{ab}\mathrm{Q}^\dagger_{bs}, \tag{25.3.4}$$

where $\mathcal{P}$ is a numerical matrix, and the matrix e is given by Eq. (25.2.9).

We can learn something about the properties of the matrix $\mathcal{P}$ by requiring that Eq. (25.3.4) be consistent with the fundamental anticommutation relation (25.2.7). Eq. (25.3.4) and its adjoint yield

$$P^{-1}\{\mathrm{Q}_{ar}, \mathrm{Q}^\dagger_{bs}\}P = \sum_{cdtu} \mathcal{P}_{rt}e_{ac}\mathcal{P}^*_{su}e_{bd}\{\mathrm{Q}^\dagger_{ct}, \mathrm{Q}_{du}\}.$$

Inserting Eq. (25.2.7), this becomes

$$\delta_{rs}\sigma^\mu_{ab}P^{-1}P_\mu P = \sum_{cdtu} \mathcal{P}_{rt}e_{ac}\mathcal{P}^*_{su}e_{bd}\delta_{tu}\sigma^\mu_{dc}P_\mu.$$

But $e\sigma_i^T e^{-1} = -\sigma_i$ and $e\sigma_0^T e^{-1} = +\sigma_0$, while $P^{-1}P_iP = -P_i$ and $P^{-1}P_0P = P_0$, so this reduces to the statement that $\mathcal{P}$ is unitary

$$\mathcal{P}\mathcal{P}^\dagger = \mathbf{1}. \tag{25.3.5}$$

The matrix $\mathcal{P}$ is to some extent arbitrary, because for any set of fermionic generators Q_{ar} that satisfies Eqs. (25.3.2) and (25.2.7), we construct another set Q'_{ar} that also satisfies Eqs. (25.3.2) and (25.2.7) by a unitary transformation

$$\mathrm{Q}'_{ar} = \sum_s \mathcal{U}_{rs}\mathrm{Q}_{as}, \qquad \mathcal{U}^\dagger = \mathcal{U}^{-1}, \tag{25.3.6}$$

so that the parity transformation rule (25.3.4) becomes

$$P^{-1}\mathrm{Q}'_{ar}P = \sum_{bs} \mathcal{P}'_{rs}e_{ab}\mathrm{Q}'^\dagger_{bs}, \tag{25.3.7}$$

where

$$\mathcal{P}' = \mathcal{U}\mathcal{P}\mathcal{U}^{-1*} = \mathcal{U}\mathcal{P}\mathcal{U}^T. \tag{25.3.8}$$

For simple supersymmetry $\mathcal{P}$ is just a $1 \cdot 1$ phase factor, and Eq. (25.3.4) reads

$$P^{-1}\mathrm{Q}_a P = \mathcal{P}\sum_b e_{ab}\mathrm{Q}^\dagger_b. \tag{25.3.9}$$

Combining this with its adjoint gives

$$P^{-2}\mathrm{Q}_a P^2 = -\mathrm{Q}_a, \tag{25.3.10}$$

independently of the value chosen for the phase factor $\mathcal{P}$. This has the striking consequence that if a boson in a supermultiplet of particles has a *real* intrinsic parity, then the fermions obtained by acting on this boson state with Q_a have *imaginary* intrinsic parities.

Because for simple supersymmetry $\mathcal{U}$ and $\mathcal{P}$ are just phase factors, it is obvious from Eq. (25.3.8) that by a suitable choice of $\mathcal{U}$, the phase factor $\mathcal{P}'$ can be made anything we like. It will be convenient to choose $\mathcal{P}' = +i$, so that Eq. (25.3.7) takes the simple form (now dropping primes)

$$P^{-1}\mathrm{Q}_a P = i\sum_b e_{ab}\mathrm{Q}_b^\dagger. \tag{25.3.11}$$

Just as for spinor field operators, the representation of space inversion is simpler if we combine the $(0,1/2)$ operators Q_a and the $(1/2,0)$ operators $\sum_b e_{ab}\mathrm{Q}_b^\dagger$ into the four-component Dirac spinor generator Q_α defined by Eq. (25.2.34). In these terms, Eq. (25.3.11) and its adjoint read

$$P^{-1}QP = i\beta Q. \tag{25.3.12}$$

(We are using the notation for Dirac matrices given in Section 5.4 and in the Preface to this volume, which gives

$$\beta = \begin{pmatrix} 0 & \mathbf{1} \\ \mathbf{1} & 0 \end{pmatrix},$$

where $\mathbf{1}$ and 0 are understood as $2\cdot 2$ submatrices.)

For extended supersymmetry it is not always possible to choose $\mathcal{U}$ so that $\mathcal{P}'$ is diagonal. However, a theorem in matrix algebra (proved in Appendix C of Chapter 2) shows that it is possible to choose $\mathcal{U}$ so that $\mathcal{P}'$ is block diagonal, in general with some of the matrices on the diagonal equal to $1\cdot 1$ submatrices that can be chosen equal to i (or any other phase factors we like), and other submatrices on the diagonal given by $2\cdot 2$ matrices that can be chosen to have the form

$$\begin{pmatrix} 0 & \exp(i\phi) \\ \exp(-i\phi) & 0 \end{pmatrix},$$

with various phases ϕ. Correspondingly, with this choice of $\mathcal{U}$ (and now dropping primes) the two-component Qs are of two types. The Qs of the first type also satisfy Eq. (25.3.11):

$$P^{-1}\mathrm{Q}_{ar}P = i\sum_b e_{ab}\mathrm{Q}_{br}^\dagger. \tag{25.3.13}$$

The two-component Qs of the second type come in pairs, which we will call Q_{as1} and Q_{as2}, with the sth pair having the parity transformation rule

$$P^{-1}\mathrm{Q}_{as1}P = e^{i\phi_s}\sum_b e_{ab}\mathrm{Q}_{bs2}^\dagger, \quad P^{-1}\mathrm{Q}_{as2}P = e^{-i\phi_s}\sum_b e_{ab}\mathrm{Q}_{bs1}^\dagger. \tag{25.3.14}$$

In particular, we now have

$$P^{-2}\mathrm{Q}_{as1}P^2 = -e^{2i\phi_s}\mathrm{Q}_{as1}, \qquad P^{-2}\mathrm{Q}_{as2}P^2 = -e^{-2i\phi_s}\mathrm{Q}_{as2}. \tag{25.3.15}$$

This shows that, unless $\phi_s = 0 \pmod{\pi}$, it is not possible to form supersymmetry generators of the first type out of linear combinations of extended supersymmetry generators of the second type.

In terms of the four-component spinors (25.2.34), the effect of the parity operator on extended supersymmetry generators of the first type is

$$P^{-1}Q_rP = i\beta Q_r, \tag{25.3.16}$$

while for generators of the second type

$$P^{-1}Q_{s1}P = \beta\gamma_5 \exp(i\gamma_5\phi_s)Q_{s2}, \quad P^{-1}Q_{s2}P = \beta\gamma_5 \exp(-i\gamma_5\phi_s)Q_{s1}. \tag{25.3.17}$$

25.4 Massless Particle Supermultiplets

Supersymmetry requires that the known particles be accompanied in irreducible representations of the supersymmetry algebra by 'sparticles': bosonic 'squarks' and 'sleptons' accompanying quarks and leptons, and fermionic 'gauginos' accompanying gauge bosons. None of these sparticles have been observed, so supersymmetry is certainly broken, with the masses of the sparticles almost certainly much larger than the quark, lepton, and gauge boson masses produced by the spontaneous breakdown of the electroweak $SU(2)\cdot U(1)$ gauge group, and hence of the same order of magnitude as the splittings within supermultiplets. Thus it is very likely that at energy scales that are large enough so that we can neglect supersymmetry breaking and these mass splittings, we can also treat the known quarks, leptons, and gauge bosons and their superpartners as massless. Therefore we will be specially interested in supermultiplets of massless particles.

Consider a state containing a single massless particle belonging to some supermultiplet. We obtain the other states in the same supermultiplet by applying operators Q_{ar} and/or $\mathrm{Q}^\dagger_{ar}$ to this state. Since Q_{ar} and $\mathrm{Q}^\dagger_{ar}$ commute with P_μ, all these states have the same value of the four-momentum. We will work in a Lorentz frame in which the four-momentum of these states is $p^1 = p^2 = 0$ and $p^3 = p^0 = E$. With this choice of four-momentum, we have

$$\sigma_\mu p^\mu = E(\sigma_0 + \sigma_3) = 2E\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \tag{25.4.1}$$

which aside from the factor $2E$ is the projection matrix onto the subspace with helicity $+1/2$. The anticommutation relation (25.2.7) therefore shows that $\{\mathrm{Q}_{(-1/2)r}, \mathrm{Q}^\dagger_{(-1/2)r}\}$ gives zero when acting on any state in a supermultiplet with this momentum, and thus so do $\mathrm{Q}_{(-1/2)r}$ and $\mathrm{Q}^\dagger_{(-1/2)r}$. We must therefore construct the states of the supermultiplet by acting only with $\mathrm{Q}_{(1/2)r}$ and $\mathrm{Q}^\dagger_{(1/2)r}$. Furthermore, we are labelling the Qs with their J_3 values, in the sense that

$$[J_3, \mathrm{Q}_{ar}] = -a\mathrm{Q}_{ar}\,, \tag{25.4.2}$$

so $\mathrm{Q}_{(1/2)r}$ and $\mathrm{Q}^\dagger_{(1/2)r}$ respectively lower and raise the helicity by 1/2.

We will first consider the case of simple supersymmetry. Consider a supermultiplet with maximum helicity $\lambda_{\max}$, and let $\|\lambda_{\max}\rangle$ be any one-particle state with this helicity and four-momentum p^μ. Then

$$\mathrm{Q}^\dagger_{\frac{1}{2}}\|\lambda_{\max}\rangle = 0\,, \tag{25.4.3}$$

while acting on this state with $Q_{1/2}$ gives a state $\|\lambda_{\rm max}-1/2\rangle$ with helicity $\lambda_{\rm max}-1/2$. We will define this state as

$$\|\lambda_{\rm max}-1/2\rangle \equiv (4E)^{-1/2} Q_{\frac{1}{2}}\|\lambda_{\rm max}\rangle\,. \tag{25.4.4}$$

The fundamental anticommutation relation (25.2.7) together with Eqs. (25.4.1) and (25.4.3) shows that this state is normalized in the same way as $\|\lambda_{\rm max}\rangle$

$$\langle\lambda_{\rm max}-1/2\|\lambda_{\rm max}-1/2\rangle = \langle\lambda_{\rm max}\|\lambda_{\rm max}\rangle\,, \tag{25.4.5}$$

and in particular this state cannot vanish. Eq. (25.2.32) shows that $Q^2_{1/2}=0$, so acting with $Q_{1/2}$ on $\|\lambda_{\rm max}-1/2\rangle$ gives zero:

$$Q_{\frac{1}{2}}\|\lambda_{\rm max}-1/2\rangle = (4E)^{-1/2} Q^2_{\frac{1}{2}}\|\lambda_{\rm max}\rangle = 0\,. \tag{25.4.6}$$

On the other hand, acting with $Q^\dagger_{1/2}$ on this state gives the state with which we started. That is,

$$\begin{aligned} Q^\dagger_{\frac{1}{2}}\|\lambda_{\rm max}-1/2\rangle &= (4E)^{-1/2} Q^\dagger_{\frac{1}{2}} Q_{\frac{1}{2}}\|\lambda_{\rm max}\rangle \\ &= (4E)^{-1/2}\{Q^\dagger_{\frac{1}{2}}, Q_{\frac{1}{2}}\}\|\lambda_{\rm max}\rangle\,, \end{aligned}$$

so that Eq. (25.4.1) and the anticommutation relation (25.2.31) yield

$$Q^\dagger_{\frac{1}{2}}\|\lambda_{\rm max}-1/2\rangle = (4E)^{1/2}\|\lambda_{\rm max}\rangle\,. \tag{25.4.7}$$

Thus the supermultiplet consists of just two states, with helicities $\lambda_{\rm max}$ and $\lambda_{\rm max}-1/2$. In the basis provided by these two states, the operators $Q_{1/2}$ and $Q^\dagger_{1/2}$ are represented by the matrices

$$q_{\frac{1}{2}} = \sqrt{4E}\begin{pmatrix}0&0\\1&0\end{pmatrix}\,, \qquad q^\dagger_{\frac{1}{2}} = \sqrt{4E}\begin{pmatrix}0&1\\0&0\end{pmatrix}\,, \tag{25.4.8}$$

while the operators $Q_{-1/2}$ and $Q^\dagger_{-1/2}$ are represented by zero.

It is worth emphasizing that this is the *only* kind of massless supermultiplet in theories with simple supersymmetry. There are no massless particles that are not accompanied with a superpartner, and none that have more than one superpartner. Of course, CPT invariance implies that for every supermultiplet of massless particles of helicities λ and $\lambda-1/2$ there must be an antimultiplet with helicities $-\lambda+1/2$ and $-\lambda$. In particular, a massless particle and antiparticle with helicities $+1/2$ and $-1/2$ must be accompanied by a massless particle and antiparticle either with helicities $+1$ and -1 or with helicities both zero.

How might the known quarks, leptons, and gauge bosons fit into this picture? We will assume that the supersymmetry generator commutes with the generators of the $SU(3)\cdot SU(2)\cdot U(1)$ gauge group[5]. The quarks and leptons belong to different representations of the gauge group from the gauge bosons, so they cannot be in the same supermultiplets. We have to conclude then that in the limit of high energy where $SU(2)\cdot U(1)$ symmetry

[5]In simple supersymmetry the generator Q_α must in any case commute with the $SU(3)\cdot SU(2)$ generators, because semi-simple algebras like $SU(3)\cdot SU(2)$ have no non-trivial one-dimensional representations.

breaking may be neglected, the massless quarks and leptons of each color and flavor are in supermultiplets with *pairs* of massless squarks and sleptons of zero helicity and the same color and flavor, while the massless gauge bosons are accompanied by massless gauginos of helicity $\pm 1/2$ comprising an adjoint representation of $SU(3) \cdot SU(2) \cdot U(1)$.

Because gravity exists we know that in addition to the particles of the standard model there must also exist a massless particle of helicity ± 2, the *graviton*. Massless particles with helicity λ having $|\lambda| > 1/2$ must couple at low momentum to conserved quantities.[6] Soft massless particles of helicity ± 1 can couple to various internal symmetry generators, soft massless particles of helicity $\pm 3/2$ can couple to the supersymmetry generators Q_a, and a soft particle of helicity ± 2 can couple to a single conserved quantity, the momentum four-vector P_μ, but there are no conserved quantities to which a soft massless particle with $|\lambda| > 2$ could couple. We conclude that the graviton cannot be in a supermultiplet with particles of helicity $\pm 5/2$, so it must be in a supermultiplet with a massless particle of helicity $\pm 3/2$, known as a *gravitino*, coupled to the supersymmetry generators themselves. The field theory of this supermultiplet is known as *supergravity*, and will be discussed in Chapter 31.

Now let us consider the case of extended supersymmetry, with N supersymmetry generators. We first note that because the $\mathrm{Q}_{(-1/2)r}$ all give zero when acting on the states of a supermultiplet (including a state obtained by letting $\mathrm{Q}_{(1/2)s}$ act on any other state of the multiplet), the central charges Z_{rs} must also annihilate any state of the multiplet. With central charges out of the picture, the supersymmetry generators $\mathrm{Q}_{(1/2)r}$ all anticommute when acting on a massless particle supermultiplet, so applying n of them to a one-particle state of maximum helicity $\lambda_{\max}$ and four-momentum p^μ gives $N!/n!(N-n)!$ one-particle states of the same four-momentum and helicity $\lambda_{\max} - n/2$, forming a rank n antisymmetric tensor representation of the $SU(N)$ R-symmetry[7] (25.2.30). The maximum value of n that gives a non-zero state is $n = N$, so the minimum helicity in a supermultiplet is given by

$$\lambda_{\min} = \lambda_{\max} - N/2\,. \tag{25.4.9}$$

If we wish to exclude massless particle helicities λ with $|\lambda| > 2$, then $\lambda_{\max} - \lambda_{\min} \leq 4$, so extended supersymmetries are allowed only with $N \leq 8$.

For $N = 8$, with helicities with $|\lambda| > 2$ excluded, there is only one possible supermultiplet, consisting of: 1 graviton with each helicity ± 2; 8 gravitinos with each helicity $\pm 3/2$; 28 gauge bosons with each helicity ± 1; 56 fermions with each helicity $\pm 1/2$; and 70 bosons with helicity zero.

Compare this with the case $N = 7$, again excluding helicities with $|\lambda| > 2$. Here there are two supermultiplets. One supermultiplet contains: 1 graviton with helicity $+2$; 7 gravitinos with helicity $+3/2$; 21 gauge bosons with helicity $+1$; 35 fermions with helicity $+1/2$; 35 bosons with helicity zero; 21 fermions with helicity $-1/2$; 7 gauge bosons with helicity -1; and 1 gravitino with helicity $-3/2$. The other is the CPT-conjugate supermultiplet, with all helicities reversed. Adding the numbers of particles in these two supermultiplets,

[6] This is discussed for the case of integer helicity in Section 13.1. The argument for half-integer helicity was given by Grisaru and Pendleton [3].

[7] The $U(1)$ part of $U(N)$ R-symmetry is often violated by quantum mechanical anomalies.

we have 1 graviton with each helicity ± 2; $7+1=8$ gravitinos with each helicity $\pm 3/2$; $21+7=28$ gauge bosons with each helicity ± 1; $35+21=56$ fermions with each helicity $\pm 1/2$; and $35+35=70$ bosons with helicity zero. Extended supergravity theories with $N=8$ and $N=7$ thus have precisely the same particle content and are in fact identical.

On the other hand, extended supergravity theories with $N \leq 6$ have just N gravitinos of each helicity $\pm 3/2$ and are therefore all distinct.

For $N \leq 4$ there is also the possibility of *global* supersymmetry theories, theories with supermultiplets that do not include gravitons or gravitinos. For global $N=4$ supersymmetry there is just one supermultiplet, containing: 1 gauge boson of each helicity ± 1; 4 fermions of each helicity $\pm 1/2$; and 6 bosons of helicity zero. This is equivalent to the global supersymmetry theory with $N=3$, which has two supermultiplets: 1 supermultiplet with one gauge boson of helicity $+1$, 3 fermions with helicity $+1/2$; 3 bosons with helicity zero; and 1 fermion with helicity $-1/2$; and the other the CPT conjugate supermultiplet with opposite helicities. Adding the numbers of particles of each helicity in these two $N=3$ supermultiplets gives the same particle content as for $N=4$ global supersymmetry. The gauge field theory with $N=4$ supersymmetry has remarkable properties, which will be discussed in Section 27.9.

For $N=2$ extended global supersymmetry there are supermultiplets of two different types, apart from those related by CPT. There are gauge supermultiplets, each containing one gauge boson of helicity $+1$, two fermions of helicity $+1/2$ forming a doublet under the $SU(2)$ R-symmetry (25.2.30), and one boson of helicity zero, together with their CPT-conjugate supermultiplets, with helicities reversed. Together each gauge supermultiplet and its antimultiplet contain one gauge boson with each helicity ± 1, an $SU(2)$ doublet of fermions of each helicity $\pm 1/2$, and two $SU(2)$ singlet bosons of helicity zero. Then there are *hypermultiplets*, which contain one fermion of each helicity $\pm 1/2$ and an $SU(2)$ doublet of bosons of helicity zero, together with its CPT-conjugate. (In quantum field theory a hypermultiplet cannot be its own antimultiplet, because then the helicity zero particles would be described by just two *real* scalar fields, which cannot form an $SU(2)$ doublet.) Of course, in the real world there would also have to be a graviton supermultiplet, containing a graviton of helicity $+2$, an $SU(2)$ doublet of gravitinos with helicity $+3/2$, and one gauge boson with helicity $+1$, together with their CPT-conjugates, with opposite helicity. Gauge theories with $N=2$ supersymmetry are constructed in Section 27.9, and explored non-perturbatively in Section 29.5.

The particle content of these supermultiplets reveals a difficulty in incorporating extended supersymmetry in realistic theories of particles at accessible energies. In all cases but one, the helicity $+1/2$ fermions belong to supermultiplets along with helicity $+1$ gauge bosons. Gauge bosons belong to the adjoint representation of the gauge group, so if the supersymmetry generators are invariant under the gauge group then the helicity $+1/2$ fermions must also belong to the adjoint representation, which is real. This is in conflict with the fact that the known quarks and leptons belong to a representation of $SU(3) \cdot SU(2) \cdot U(1)$ which is *chiral*—that is, for which the helicity $+1/2$ fermions belong to a complex representation, which is then necessarily different from the representation furnished by their CPT-conjugates, the helicity $-1/2$ fermions. The one exception, where helicity $+1/2$ fermions are not in a supermultiplet with gauge bosons, is the $N = 2$ hypermultiplet discussed above. But in this case particles of both helicities $+1/2$ and $-1/2$ are in the same supermultiplet, and therefore must transform the same under any gauge transformations that leave the supersymmetry generators invariant. They may belong to a complex representation of this gauge group, but then the CPT-conjugate of this hypermultiplet belongs to the complex-conjugate representation, and the fermions of each helicity then belong to the sum of the two representations, which is real, again in conflict with the chiral nature of the known quarks and leptons.

In contrast, for simple supersymmetry there are supermultiplets containing just helicity $+1/2$ and helicity zero, which may be in a complex representation of the gauge group, distinct from the representation furnished by the CPT-conjugate supermultiplet. Here there is no conflict with chirality. For this reason, most discussions of supersymmetry as a symmetry that remains unbroken at accessible energies have focused on simple rather than extended supersymmetry.

25.5 Massive Particle Supermultiplets

Although the known quarks, leptons, and gauge bosons and their superpartners may probably be treated as massless at energies where supersymmetry breaking is negligible, this is not necessarily true for other particles, including the extra gauge bosons of large mass required by theories that unify the strong and electroweak interactions. Also, ever since the Wess–Zumino model, massive particle theories have been useful test cases for studying supersymmetry theories. It will therefore be worthwhile for us briefly to consider the implications of unbroken supersymmetry for massive particles.

As in the previous section, we obtain the various one-particle states in a supermultiplet by acting on any one of them with the operators Q_{ar} and $\mathrm{Q}^{\dagger}_{ar}$, and all of these states have the same four-momentum. Unlike the case of zero mass, for mass $M > 0$ we may now take this to be the four-momentum of a particle at rest, with $p^i = 0$ for $i = 1, 2, 3$ and $p^0 = M$. In this frame of reference, we have

$$\sigma_\mu p^\mu = M\sigma_0 = M \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} . \tag{25.5.1}$$

Thus, acting on any state $\| \, \rangle$ in a supermultiplet with this four-momentum, the anticommutation relation (25.2.7) yields

$$\{\mathrm{Q}_{ar}, \mathrm{Q}^{\dagger}_{bs}\} \| \, \rangle = 2M\delta_{ab}\delta_{rs} \| \, \rangle . \tag{25.5.2}$$

In contrast with the case of zero mass, here no component of Q_{ar} or $\mathrm{Q}^{\dagger}_{ar}$ can vanish on the whole multiplet, so we have two sets of raising and lowering operators: both $\mathrm{Q}_{(1/2)r}$ and $\mathrm{Q}^{\dagger}_{(-1/2)r}$ lower the spin 3-component by 1/2, while both $\mathrm{Q}_{(-1/2)r}$ and $\mathrm{Q}^{\dagger}_{(1/2)r}$ raise the spin 3-component by 1/2. However, as we shall see, for extended supersymmetry it is possible for certain linear combinations of the Qs and $\mathrm{Q}^{\dagger}$s to vanish.

We will first consider the case of simple supersymmetry. By using the supersymmetry algebra (25.2.31), (25.2.32), we shall show that the general massive supermultiplet consists of a particle of spin $j+1/2$, a pair of particles of spin j, and a particle of spin $j-1/2$. Where parity is conserved, the particles of spin $j \pm 1/2$ have equal intrinsic parity, given by some phase η, while the two particles of spin j have parities $+i\eta$ and $-i\eta$. Here j is any integer or half-integer greater than zero. There is also a collapsed supermultiplet, consisting of two particles of spin zero and a particle of spin 1/2. When parity is conserved the particles of spin zero have parities $i\eta$ and $-i\eta$, where η is the parity of the particle of spin 1/2.

Here is the proof. We first show that any supermultiplet will contain at least one spin multiplet of states $\|j,\sigma\rangle$ with spin 3-component σ running by unit steps from $-j$ to $+j$, having the special property that, for all such σ and for $a = \pm 1/2$,

$$\mathrm{Q}_a \|j,\sigma\rangle = 0. \tag{25.5.3}$$

Starting with any non-zero state $\|\psi\rangle$ in the supermultiplet, we can define non-zero states

$$\|\psi'\rangle \equiv \begin{cases} (2M)^{-1/2}\mathrm{Q}_{1/2}\|\psi\rangle, & \mathrm{Q}_{1/2}\|\psi\rangle \neq 0, \\ \|\psi\rangle, & \mathrm{Q}_{1/2}\|\psi\rangle = 0, \end{cases}$$

and

$$\|\psi''\rangle \equiv \begin{cases} (2M)^{-1/2}\mathrm{Q}_{-1/2}\|\psi'\rangle, & \mathrm{Q}_{-1/2}\|\psi'\rangle \neq 0, \\ \|\psi'\rangle, & \mathrm{Q}_{-1/2}\|\psi'\rangle = 0. \end{cases}$$

Because the Q_a anticommute, $\mathrm{Q}_{1/2}\|\psi'\rangle = 0$, and so $\mathrm{Q}_a\|\psi''\rangle = 0$ for $a = \pm 1/2$. If any state $\|\psi''\rangle$ satisfies the condition that $\mathrm{Q}_a\|\psi''\rangle = 0$, then so does $U(R)\|\psi''\rangle$, where $U(R)$ is the unitary operator representing an arbitrary spatial rotation. It follows that the states that satisfy this condition may be decomposed into complete spin multiplets $\|j,\sigma\rangle$, satisfying condition (25.5.3).

Now focus on any one of these spin multiplets satisfying Eq. (25.5.3), normalized so that

$$\langle j,\sigma' \| j,\sigma\rangle = \delta_{\sigma'\sigma}. \tag{25.5.4}$$

For $j > 0$, by applying the spin 1/2 operators[8] $\mathrm{Q}_a^\dagger$ to these states we can construct states of spin $j \pm 1/2$:

$$\|j \pm 1/2, \sigma\rangle = \frac{1}{\sqrt{2M}} \sum_a C_{\frac{1}{2}j}(j \pm 1/2, \sigma; a, \sigma - a)\mathrm{Q}_a^\dagger \|j, \sigma - a\rangle, \tag{25.5.5}$$

where $C_{jj'}(j'', \sigma''; \sigma, \sigma')$ is the conventional Clebsch–Gordan coefficient for coupling spins j and j' with 3-components σ and σ' to make spin j'' with 3-component σ''. Using Eqs. (25.5.2)–(25.5.5) and the orthonormality properties of the Clebsch–Gordan coefficients, we can show that these states are properly normalized:

$$\langle j \pm 1/2, \sigma \| j \pm 1/2, \sigma'\rangle = \delta_{\sigma\sigma'}, \qquad \langle j \pm 1/2, \sigma \| j \mp 1/2, \sigma'\rangle = 0, \tag{25.5.6}$$

[8]Since Q_a transforms under rotations like a field that destroys a particle of spin 1/2 and spin 3-component a, it is $\mathrm{Q}_a^\dagger$ that transforms like a field that creates such a particle, and hence transforms like the particle itself. To put this formally, $[J_i, \mathrm{Q}_a] = -\sum_b \frac{1}{2}(\sigma_i)_{ab}\mathrm{Q}_b$, so $[J_i, \mathrm{Q}_a^\dagger] = \sum_b \frac{1}{2}(\sigma_i)_{ba}\mathrm{Q}_b^\dagger$, which may be compared with the transformation property of a spin 1/2 particle, $J_i\|a\rangle = \sum_b \frac{1}{2}(\sigma_i)_{ba}\|b\rangle$.

so none of the states $\|j \pm 1/2, \sigma\rangle$ can vanish. The only exception is for $j = 0$, in which case of course there is no state $\|j - 1/2, \sigma\rangle$. We can also construct other states by applying *two* $\mathrm{Q}^\dagger$s to $\|j, \sigma\rangle$. Since each $\mathrm{Q}^\dagger_a$ anticommutes with itself, the only such non-zero states are formed by applying the operator $\mathrm{Q}^\dagger_{1/2}\mathrm{Q}^\dagger_{-1/2} = -\mathrm{Q}^\dagger_{-1/2}\mathrm{Q}^\dagger_{1/2}$. This operator may be written as $\frac{1}{2}e_{ab}\mathrm{Q}^\dagger_a\mathrm{Q}^\dagger_b$, which shows that it is rotationally invariant, so this gives a second spin multiplet with spin j:

$$\|j, \sigma\rangle^{\mathrm{b}} = \frac{1}{2M}\mathrm{Q}^\dagger_{1/2}\mathrm{Q}^\dagger_{-1/2}\|j, \sigma\rangle, \tag{25.5.7}$$

which is distinguished from $\|j, \sigma\rangle$ by the fact that instead of Eq. (25.5.3), we have

$$\mathrm{Q}^\dagger_a\|j, \sigma\rangle^{\mathrm{b}} = 0. \tag{25.5.8}$$

Again using Eqs. (25.5.2)–(25.5.4), we find that these are also normalized states:

$${}^{\mathrm{b}}\langle j, \sigma'\|j, \sigma\rangle^{\mathrm{b}} = \delta_{\sigma'\sigma}, \qquad \langle j, \sigma'\|j, \sigma\rangle^{\mathrm{b}} = 0. \tag{25.5.9}$$

It is then easy to show that the states constructed so far form a complete representation of the supersymmetry algebra. The orthonormality property of the Clebsch–Gordan coefficients allows us to rewrite Eq. (25.5.5) as

$$\mathrm{Q}^\dagger_a\|j, \sigma\rangle = \sqrt{2M}\sum_{\pm} C_{\frac{1}{2}j}(j \pm 1/2, \sigma + a; a, \sigma)\|j \pm 1/2, \sigma + a\rangle. \tag{25.5.10}$$

Also, Eq. (25.5.2) shows that, for any state $\|\ \rangle$ in the supermultiplet,

$$[\mathrm{Q}_a, \mathrm{Q}^\dagger_{1/2}\mathrm{Q}^\dagger_{-1/2}]\|\ \rangle = 2M\sum_b e_{ab}\mathrm{Q}^\dagger_b\|\ \rangle, \tag{25.5.11}$$

so Eqs. (25.5.7) and (25.5.3) give

$$\begin{aligned} \mathrm{Q}_a\|j, \sigma\rangle^{\mathrm{b}} &= \sum_b e_{ab}\mathrm{Q}^\dagger_b\|j, \sigma\rangle \\ &= \sqrt{2M}\sum_b e_{ab}\sum_{\pm} C_{\frac{1}{2}j}(j \pm 1/2, \sigma + b; b, \sigma)\|j \pm 1/2, \sigma + b\rangle. \end{aligned} \tag{25.5.12}$$

From Eqs. (25.5.2), (25.5.3), and (25.5.5) we have

$$\mathrm{Q}_a\|j \pm 1/2, \sigma\rangle = \sqrt{2M}\, C_{\frac{1}{2}j}(j \pm 1/2, \sigma; a, \sigma - a)\|j, \sigma - a\rangle, \tag{25.5.13}$$

while Eqs. (25.5.5), (25.2.31), and (25.5.7) yield

$$\mathrm{Q}^\dagger_a\|j \pm 1/2, \sigma\rangle = \sqrt{2M}\sum_b e_{ab}C_{\frac{1}{2}j}(j \pm 1/2, \sigma; b, \sigma - b)\|j, \sigma - b\rangle^{\mathrm{b}}. \tag{25.5.14}$$

Eqs. (25.5.3), (25.5.8), (25.5.10), and (25.5.12)–(25.5.14) give the action of the Qs and $\mathrm{Q}^\dagger$s on all the states of the supermultiplet.

For $j = 0$ we have the collapsed supermultiplet: Eqs. (25.5.3), (25.5.8), (25.5.10), and (25.5.12)–(25.5.14) become

$$
\begin{aligned}
&\mathrm{Q}_a\|0,0\rangle = 0, && \mathrm{Q}_a^\dagger\|0,0\rangle^{\mathrm{b}} = 0,\\
&\mathrm{Q}_a^\dagger\|0,0\rangle = \sqrt{2M}\|1/2,a\rangle, && \mathrm{Q}_a\|0,0\rangle^{\mathrm{b}} = \sqrt{2M}\sum_b e_{ab}\|1/2,b\rangle,\\
&\mathrm{Q}_a\|1/2,b\rangle = \sqrt{2M}\delta_{ab}\|0,0\rangle, && \mathrm{Q}_a^\dagger\|1/2,b\rangle = \sqrt{2M}e_{ab}\|0,0\rangle^{\mathrm{b}}.
\end{aligned}
\tag{25.5.15}
$$

Now suppose that parity is conserved. Recall that the phase of the supersymmetry generator may be chosen so that the action of the parity operator on these generators is given by Eq. (25.3.13). Then $\mathrm{Q}_a^\dagger$ acting on $P\|j,\sigma\rangle$ is a linear combination of the states $P\mathrm{Q}_a\|j,\sigma\rangle$, which vanish, and since $P\|j,\sigma\rangle$ has the same rotation properties as $\|j,\sigma\rangle^{\mathrm{b}}$, it must simply be proportional to it

$$
P\|j,\sigma\rangle = -\eta\|j,\sigma\rangle^{\mathrm{b}}. \tag{25.5.16}
$$

Since P is unitary, η is a phase factor, with $|\eta| = 1$. A corresponding argument shows that $P\|j,\sigma\rangle^{\mathrm{b}}$ is proportional to $\|j,\sigma\rangle$. To find the proportionality coefficient, we note that

$$
\begin{aligned}
P\|j,\sigma\rangle^{\mathrm{b}} &= (2M)^{-1}P\mathrm{Q}_{1/2}^\dagger\mathrm{Q}_{-1/2}^\dagger\|j,\sigma\rangle = -\eta(2M)^{-1}\mathrm{Q}_{-1/2}\mathrm{Q}_{1/2}\|j,\sigma\rangle^{\mathrm{b}}\\
&= -\eta(2M)^{-2}\mathrm{Q}_{-1/2}\mathrm{Q}_{1/2}\mathrm{Q}_{1/2}^\dagger\mathrm{Q}_{-1/2}^\dagger\|j,\sigma\rangle = -\eta\|j,\sigma\rangle.
\end{aligned}
$$

We can then define states of spin j

$$
\|j,\sigma\rangle^{\pm} \equiv \frac{1}{\sqrt{2}}\left(\|j,\sigma\rangle \pm i\|j,\sigma\rangle^{\mathrm{b}}\right), \tag{25.5.17}
$$

with definite parity

$$
P\|j,\sigma\rangle^{\pm} = \pm i\eta\|j,\sigma\rangle^{\pm}. \tag{25.5.18}
$$

Finally, applying the parity operator to Eq. (25.5.5) and using Eqs. (25.3.13) and (25.5.16) gives

$$
P\|j\pm 1/2,\sigma\rangle = -\frac{\eta}{\sqrt{2M}}\sum_a C_{\frac{1}{2}j}(j\pm 1/2,\sigma;a,\sigma-a)\sum_b e_{ab}\mathrm{Q}_b\|j,\sigma-a\rangle^{\mathrm{b}}.
$$

Eq. (25.5.12) and the orthonormality property of the Clebsch–Gordan coefficients then yield

$$
P\|j\pm 1/2,\sigma\rangle = \eta\|j\pm 1/2,\sigma\rangle, \tag{25.5.19}
$$

as was to be shown.

We now turn briefly to the case of extended supersymmetry, with N supersymmetry generators. As mentioned in the previous section, there can be no massless particle with a non-vanishing eigenvalue for any central charge. We can go further and show that the eigenvalues of the central charge operators set a lower bound on the mass of any supermultiplet.

Because the central charges Z_{rs} and $Z_{rs}^\dagger$ commute with each other and with P_μ, one-particle states can be chosen to be eigenstates of all the central charges as well as of P_μ, and because the central charges commute with the Q_{ar} and $\mathrm{Q}_{ar}^\dagger$, all states in a supermultiplet have the same eigenvalues.

To derive an inequality relating the mass M of a supermultiplet and the eigenvalues of the central charges on this multiplet, we use the anticommutation relations (25.2.7) and (25.2.8) to write

$$\sum_{ar}\left\{\left(\mathrm{Q}_{ar}-\sum_{bs}e_{ab}U_{rs}\mathrm{Q}_{bs}^\dagger\right),\left(\mathrm{Q}_{ar}^\dagger-\sum_{ct}e_{ac}U_{rt}^*\mathrm{Q}_{ct}\right)\right\}$$
$$=8NP^0-2\,\mathrm{Tr}\left(ZU^\dagger+UZ^\dagger\right), \tag{25.5.20}$$

where U_{rs} is an arbitrary $N\cdot N$ unitary matrix. The left-hand side is a positive-definite operator, so by letting this act on the states of the supermultiplet at rest, we find

$$M\geq\frac{1}{4N}\,\mathrm{Tr}\left(ZU^\dagger+UZ^\dagger\right), \tag{25.5.21}$$

where now Z_{rs} denotes the values of the central charges for the supermultiplet of mass M. The polar decomposition theorem tells us that any square matrix Z may be written as HV, where H is a positive Hermitian matrix and V is unitary. We can obtain a useful inequality (which is in fact optimal) by setting $U=V$, in which case Eq. (25.5.21) becomes

$$M\geq\frac{1}{2N}\,\mathrm{Tr}\,H=\frac{1}{2N}\,\mathrm{Tr}\sqrt{Z^\dagger Z}. \tag{25.5.22}$$

States for which M equals the minimum value allowed by this inequality are known as BPS states, by analogy with the Bogomol'nyi–Prasad–Sommerfeld magnetic monopole configurations discussed in Section 23.3, whose mass also equals a lower bound on general monopole masses. In fact, this is more than an analogy; we will see in Section 27.9 that the lower bound on monopole masses in gauge theories with extended supersymmetry is a special case of the lower bound (25.5.22).

As we can see from this derivation of Eq. (25.5.22), for BPS supermultiplets the operator $\mathrm{Q}_{ar}-\sum_{bs}e_{ab}U_{rs}\mathrm{Q}_{bs}^\dagger$ gives zero when acting on any state of a supermultiplet, so there are only N independent helicity-lowering operators $\mathrm{Q}_{(1/2)r}$ and only N independent helicity-raising operators $\mathrm{Q}_{(-1/2)r}$, just as in the case of massless supermultiplets. This leads to smaller supermultiplets than would be found in the general case.

For instance, for $N=2$ supersymmetry the central charge is given by a single complex number[9]

$$Z=\begin{pmatrix}0 & Z_{12}\\ -Z_{12} & 0\end{pmatrix}. \tag{25.5.23}$$

The inequality (25.5.22) here reads

$$M\geq|Z_{12}|/2. \tag{25.5.24}$$

Where $M=|Z_{12}|/2$, the helicity contents of the massive particle supermultiplets are the same as for those of zero mass: there are gauge supermultiplets, consisting of one particle

[9] In some articles on $N=2$ supersymmetry the central charge Z is what we would call $Z/(2\sqrt{2})$.

of spin 1, an $SU(2)$ R-symmetry doublet of spin 1/2, and one particle of spin 0 (the other helicity zero state belonging to the spin one particle), and hypermultiplets, consisting of one particle of spin 1/2 and an $SU(2)$ R-symmetry doublet of spin 0. These are sometimes called 'short' supermultiplets, to distinguish them from the larger supermultiplets encountered when $M > |Z_{12}|/2$.

Problems

1. Find a set of $2 \cdot 2$ matrices that form a graded Lie algebra containing fermionic as well as bosonic generators.

2. Following the approach of Haag, Lopuszanski, and Sohnius, derive the form of the most general symmetry superalgebra in $2+1$ spacetime dimensions. (Hint: With the generators of the Lorentz group in $2+1$ spacetime dimensions labelled

$$A_1 = -iJ_{10}, \qquad A_2 = -iJ_{20}, \qquad A_3 = J_{12},$$

the commutation relations of the Poincaré algebra are

$$[A_i, A_j] = i \sum_k \epsilon_{ijk} A_k,$$

so the representations of the homogeneous Lorentz group in $2+1$ spacetime dimensions are labelled with a single positive integer or half-integer A.) Assume that the conditions for the Coleman–Mandula theorem are satisfied here.

3. Suppose that there were no massless particles with helicity greater than $+3/2$ or less than $-3/2$. Find the most general massless particle multiplets for $N = 6$ extended supersymmetry and (using CPT symmetry) $N = 5$ extended supersymmetry. What does the comparison between the particle content you find suggest about these two extended supersymmetries?

4. What are the possible parities of the particles in the short supermultiplets of extended $N = 2$ supersymmetry?

References

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